

3.3

Statements and Functions

Building blocks

STATEMENTS

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Each instruction we give the computer is a **statement** - executed top to bottom, one after another.



```
int age { 42 }; // a statement that declares a variable and initializes it
greetPerson("Grace"); // a statement that calls a function
return 0; // a statement that ends the program
```

FUNCTIONS

A FUNCTION

A named, reusable block of statements - write the logic once, call it as many times as you like.



```
void greetPerson(std::string name) {  
    std::println("Hello, {}!", name);  
}
```

name is a **parameter** - a placeholder filled in by whoever calls it.

TWO KINDS OF ERRORS

COMPILE-TIME VS. RUNTIME

Compile-time error

The program won't even build.



```
greetPersonn("Ada");  
// typo: no such function
```

Runtime error

It builds, but breaks while running.



```
int trouble = numerator / 0;  
// compiles fine, crashes when run
```

START THE DEBUGGING HABIT NOW

Breakpoint inside `greetPerson` and inspect `name` - get comfortable doing this **before** things get complicated.

